

$$\begin{bmatrix} 1 & 3 & f \\ 0 & d-3c & g \end{bmatrix} \quad \text{Row 1} = \text{Row 1} - \frac{3}{d-3c} \times \text{Row 2} \quad \begin{bmatrix} 1 & 0 & f - \frac{3g}{d-3c} \\ 0 & d-3c & g \end{bmatrix}$$

It's obvious that $d-3c \neq 0$, i.e. $d \neq 3c$

$$\therefore x_1 = f - \frac{3g}{d-3c} \quad x_2 = \frac{g}{d-3c}$$

For the equations to be consistent, for all f and g , $d \neq 3c$ is a sufficient condition.

28) Suppose a, b, c and d are constants such that a is not zero and the system below is consistent for all possible values of f and g . What can you say about the numbers a, b, c and d ? Justify your answer.

$$\begin{aligned} ax_1 + bx_2 &= f \\ cx_1 + dx_2 &= g \end{aligned}$$

Soln: Augmented matrix: $\begin{bmatrix} a & b & f \\ c & d & g \end{bmatrix}$. Row 1 = Row 1 $\times \frac{1}{a}$ $a \neq 0$

$$\begin{bmatrix} 1 & b/a & f/a \\ c & d & g \end{bmatrix}$$

$$\text{Row 2} = \text{Row 2} - c \times \text{Row 1}$$

$$\begin{bmatrix} 1 & b/a & f/a \\ 0 & d - \frac{bc}{a} & g - \frac{fc}{a} \end{bmatrix}$$

The corresponding equations are:

$$\begin{aligned} x_1 + \frac{b}{a}x_2 &= \frac{f}{a} \quad (1) \\ (d - \frac{bc}{a})x_2 &= g - \frac{fc}{a} \quad (2) \end{aligned}$$

$$x_2 = \frac{g - \frac{fc}{a}}{d - \frac{bc}{a}} = \frac{ga - fc}{da - bc} \quad da - bc \neq 0 \Rightarrow \boxed{ad \neq bc}$$

$$x_1 + \frac{b}{a} \left(\frac{ga - fc}{da - bc} \right) = \frac{f}{a} \Rightarrow x_1 + \frac{bga - bfc}{da^2 - abc} = \frac{f}{a} \Rightarrow x_1 = \frac{f}{a} - \frac{bga - bfc}{da^2 - abc}$$

$$x_1 = \frac{fda - fca - bga + bfc}{a(da - bc)} = \frac{af(d - bc) - bga + bfc}{a(da - bc)} = \frac{fd - bg}{da - bc}$$

For the system to be consistent for all possible values of g and f , $a \neq 0$ and $ad \neq bc$ are the two sufficient conditions $\square$